



CSA0501-DATABASE MANAGEMENT SYSTEMS

LAB EXPERIMENTS

DATE: 10/07/2026

EXPERIMENT 1 -- DDL Commands (PRIMARY, FOREIGN KEY, UNIQUE, CHECK)

AIM: To implement and demonstrate the use of DDL commands with constraints such as PRIMARY KEY, FOREIGN KEY, UNIQUE, and CHECK constraints on the student_cgpa_details table using MySQL.

PROCEDURE:

1. Select the required database using the USE command.
2. Create the student_cgpa_details table with required attributes.
3. Apply PRIMARY KEY constraint on student_id to uniquely identify each student.
4. Apply UNIQUE constraint on student_name to avoid duplicate names.
5. Apply CHECK constraint on CGPA and semester values to maintain valid data.
6. Create a department table and establish a relationship using a FOREIGN KEY constraint.
7. Verify the applied constraints using DESC and SHOW CREATE TABLE commands.
8. Display the final table details and confirm successful execution.

PROGRAM:

1. TABLE

```
USE student_cgpa;
```

```
SELECT * FROM student_cgpa_details;
```

OUTPUT:

The screenshot shows the MySQL Workbench interface. At the top, the 'Result Grid' displays the contents of the 'student_cgpa_details' table. Below this, the 'Output' pane shows the execution of three SQL commands: 'USE student_cgpa;', 'USE student_cgpa;', and 'SELECT * FROM student_cgpa_details LIMIT 0, 1000'. The output for each command is shown in a table with columns for #, Time, Action, Message, and Duration / Fetch.

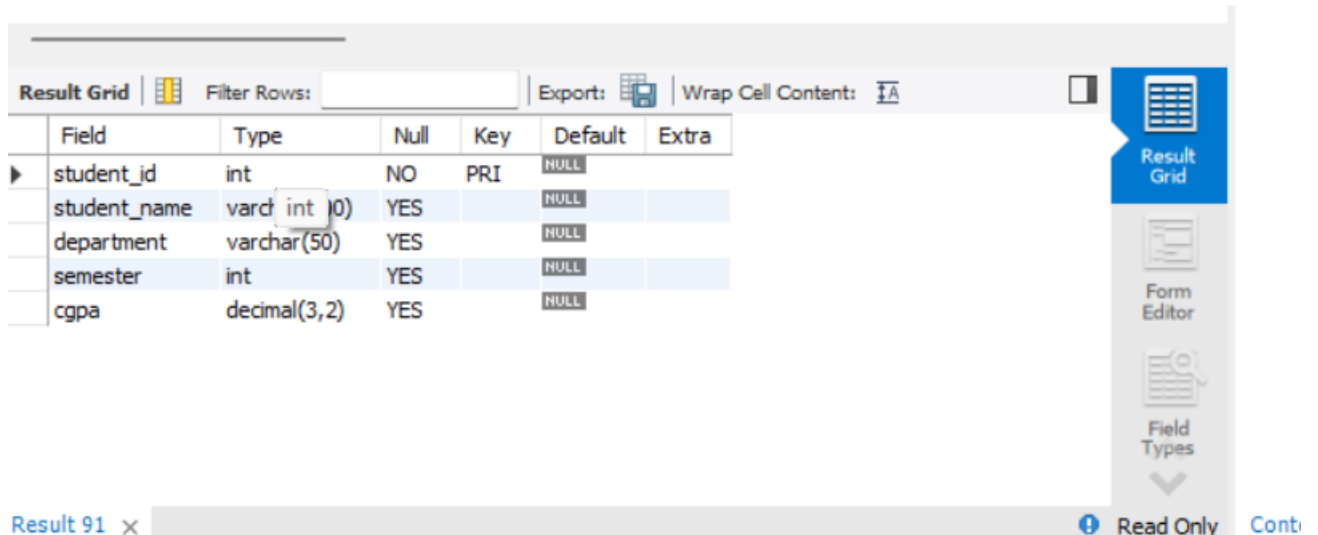
student_id	student_name	department	semester	cgpa
101	Deepthi	CSE	5	8.75
102	Anu	CSE	5	9.10
103	Rahul	ECE	5	8.50
104	Priya	IT	5	9.25
NULL	NULL	NULL	NULL	NULL

#	Time	Action	Message	Duration / Fetch
1	10:32:00	USE student_cgpa	0 row(s) affected	0.000 sec
2	10:32:14	USE student_cgpa	0 row(s) affected	0.000 sec
3	10:32:14	SELECT * FROM student_cgpa_details LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

2. PRIMARY KEY

```
DESC student_cgpa_details;  
ALTER TABLE student_cgpa_details  
ADD PRIMARY KEY(student_id);
```

OUTPUT:



The screenshot shows a 'Result Grid' window in a database management tool. The table 'student_cgpa_details' has the following structure:

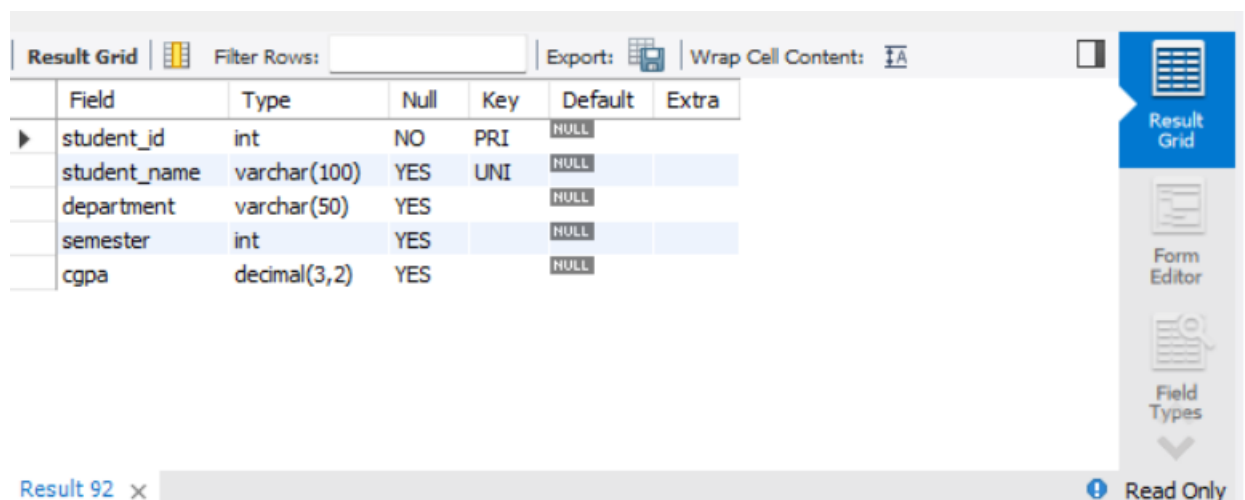
Field	Type	Null	Key	Default	Extra
student_id	int	NO	PRI	NULL	
student_name	varchar(100)	YES		NULL	
department	varchar(50)	YES		NULL	
semester	int	YES		NULL	
cgpa	decimal(3,2)	YES		NULL	

The interface includes a 'Filter Rows' input, 'Export' and 'Wrap Cell Content' buttons, and a sidebar with 'Result Grid', 'Form Editor', and 'Field Types' options. The status bar at the bottom indicates 'Result 91' and 'Read Only'.

2. UNIQUE KEY:

```
DESC student_cgpa_details;  
ALTER TABLE student_cgpa_details  
ADD CONSTRAINT unique_student_name  
UNIQUE(student_name);
```

OUTPUT:



The screenshot shows a 'Result Grid' window in a database management tool. The table 'student_cgpa_details' has the following structure:

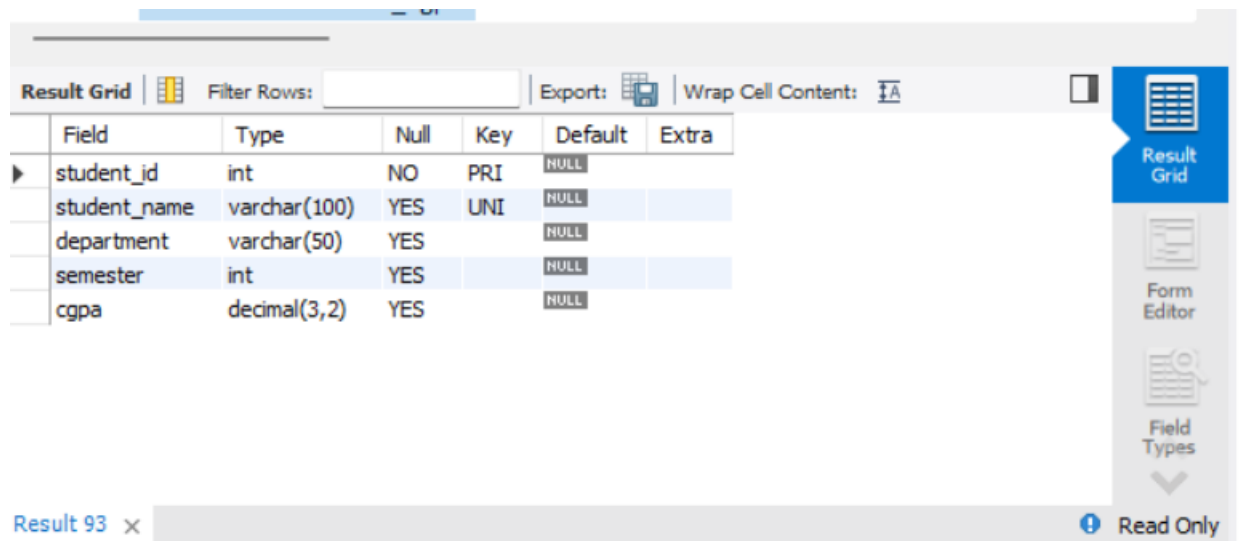
Field	Type	Null	Key	Default	Extra
student_id	int	NO	PRI	NULL	
student_name	varchar(100)	YES	UNI	NULL	
department	varchar(50)	YES		NULL	
semester	int	YES		NULL	
cgpa	decimal(3,2)	YES		NULL	

The interface includes a 'Filter Rows' input, 'Export' and 'Wrap Cell Content' buttons, and a sidebar with 'Result Grid', 'Form Editor', and 'Field Types' options. The status bar at the bottom indicates 'Result 92' and 'Read Only'.

3. CHECK CONSTRAINT:

```
DESC student_cgpa_details;  
ALTER TABLE student_cgpa_details  
ADD CONSTRAINT chk_cgpa  
CHECK(cgpa >= 0 AND cgpa <= 10);
```

OUTPUT:



Field	Type	Null	Key	Default	Extra
student_id	int	NO	PRI	NULL	
student_name	varchar(100)	YES	UNI	NULL	
department	varchar(50)	YES		NULL	
semester	int	YES		NULL	
cgpa	decimal(3,2)	YES		NULL	

4. FOREIGN KEY CONSTRAINT:

```
ALTER TABLE student_cgpa_details  
ADD CONSTRAINT fk_department  
FOREIGN KEY(dept_id)  
REFERENCES department_details(dept_id);
```

OUTPUT:

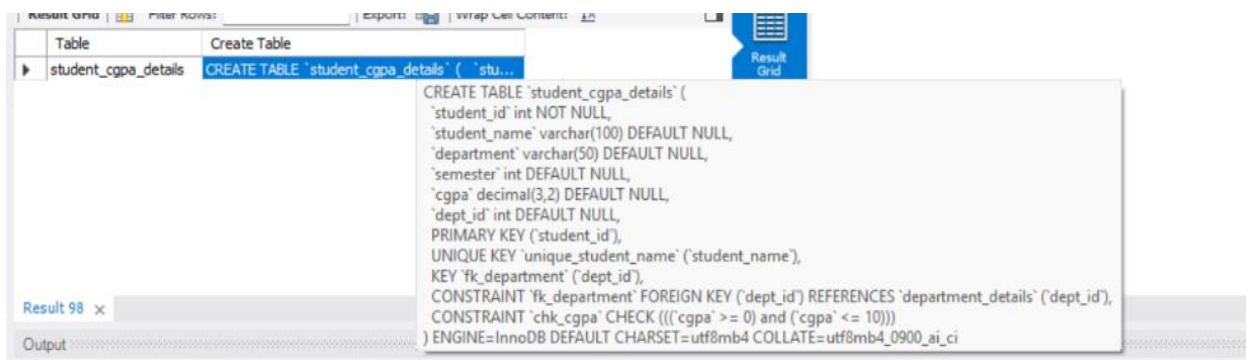


Table	Create Table
student_cgpa_details	<pre>CREATE TABLE 'student_cgpa_details' ('stu... 'student_id' int NOT NULL, 'student_name' varchar(100) DEFAULT NULL, 'department' varchar(50) DEFAULT NULL, 'semester' int DEFAULT NULL, 'cgpa' decimal(3,2) DEFAULT NULL, 'dept_id' int DEFAULT NULL, PRIMARY KEY ('student_id'), UNIQUE KEY 'unique_student_name' ('student_name'), KEY 'fk_department' ('dept_id'), CONSTRAINT 'fk_department' FOREIGN KEY ('dept_id') REFERENCES 'department_details' ('dept_id'), CONSTRAINT 'chk_cgpa' CHECK (((('cgpa' >= 0) and ('cgpa' <= 10))))) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci</pre>

RESULT: DDL Commands with Primary, Foreign, Unique, Check constraints are updated and verified.